

PHY202: Problem Set 2

1. Evaluate the integral $\int_0^{1+i} (z^2 - z) dz$ along the following two paths: (i) along the line $y = x$, (ii) along line segments $0 \rightarrow 1$ and $1 \rightarrow 1 + i$. Verify your answer using Cauchy's theorem.

2. Evaluate the integral $\oint_C \frac{\sin z}{2z - \pi} dz$, where C is: (i) the circle $|z| = 1$, (ii) $|z| = 2$.

[Hint: think about Cauchy's integral formula]

3. For the following functions, write down the Laurent's series about the indicated points and find the residue:

(i) $f(z) = \frac{1}{z(z+1)}$, about $z = 0$,

(ii) $f(z) = \frac{\sin z}{z^4}$, about $z = 0$,

(iii) $f(z) = \frac{\sin(\pi z)}{4z^2 - 1}$, $z = 1/2$.

4. Evaluate the integral $\int_0^\pi \frac{\sin^2 \theta \, d\theta}{13 - 12 \cos \theta}$ using contour integration.

[A simple method to find the residue in case of simple poles is as follows: If $f(z)$ has a simple pole at $z = z_0$, then residue of $f(z)$ at $z = z_0$ can be obtained by multiplying $f(z)$ by $(z - z_0)$ and evaluating the result at $z = z_0$.]